Dense Exponentials Do Not Force a Uniform Algebra to Be Trivial

Literature-status note for arXiv:math/0308226

9 August 2026

Status: literature already answered (negative). This is an exact later-literature resolution, not a new run proof.

Original conjecture

In Chapter 5 of *Extensions of Normed Algebras*, PDF page 53, Dawson asks whether the following condition characterizes trivial uniform algebras [1]:

$$\overline{\exp A} = A \qquad \Longrightarrow \qquad A = C(X).$$

Equivalently: must a uniform algebra whose exponentials are dense be trivial? The thesis reports that no counterexample was then known and separately highlights the natural metrizable case.

Decisive later answer

Izzo restates the exact question as Question 1.3 and says explicitly that the paper answers it [2]. Theorem 1.6 asserts that there exists a nontrivial uniform algebra A for which

$$\exp A = \{\exp a : a \in A\} \quad \text{is dense in } A.$$

The stronger Theorem 1.7 constructs a compact polynomially convex set $X \subset \mathbb{C}^2$ of topological dimension one such that $P(X)$ is a nontrivial strongly regular uniform algebra, is Dirichlet on X, and has dense exponentials. The paper also proves that X is the maximal ideal space of $P(X)$. Therefore the conjectured implication is false.

The supporting authors knew they were answering this problem: the abstract calls it a 17-year-old question of Dales and Feinstein, and the introduction labels and answers the dense-exponentials question directly.

Scope left open

The counterexample has compact metrizable, simply coconnected, one-dimensional maximal ideal space. It is not locally connected. Thus the counterexample settles the general conjecture, including the broad "natural and metrizable" formulation, but it does not settle stronger variants imposing local connectedness or requiring maximal ideal space $[0, 1]$. Izzo explicitly records related planar and locally connected questions as open.

Search evidence

The exact phrase "dense exponential group" and the formula $\overline{\exp A} = A$ were searched in the run indexes, local arXiv corpus, and bounded web search. The local source for arXiv:2403.19583 contains the explicit Question 1.3, Theorems 1.6 and 1.7, and the construction's proof of nontriviality and dense exponentials. No originality claim is made here.

References

[1] T. W. Dawson, *Extensions of Normed Algebras*, Ph.D. thesis, arXiv:math/0308226 (2003), Chapter 5, PDF page 53.

[2] A. J. Izzo, *A nontrivial uniform algebra Dirichlet on its maximal ideal space*, arXiv:2403.19583, Question 1.3 and Theorems 1.6–1.7.